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FEDERAL SPECIFICATION

PRESS BRAKE, POWER OPERATED, MECHANICAL, SHEET AND PLATE

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers power operated, mechanically driven, sheet and plate press brakes of the open throat, gap design in a range from 15 to 1000 ton capacity.

1.2 Classification. The press brakes covered by this specification shall be of the following classes and of the sizes shown in table I. The class and size required shall be as specified (see 6.2.1). The size is a combination of the bottom of the stroke tonnage rating and the maximum length of sheet or plate that the press brake is capable of bending.

Class 1 - Single speed (see 3.2.1).
Class 2 - Two speed (see 3.2.2).
Class 3 - Variable speed (see 3.2.3).
Class 4 - Automatic cycle (see 3.2.4).

2. APPLICABLE DOCUMENTS

2.1 Government Publications. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standards

FED-STD-H28 - Screw thread Standards for Federal Services.
FED-STD-123 - Marking for Shipment (Civil Agencies).
FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

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(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

Military Standard

MIL-STD-461 - Electromagnetic Emmision and Susceptibility Requirements for the Control of Electromagnetic Interference.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Federal Regulation

Occupational Health and Safety Administration (OSHA) 29 CFR 1910.212 General Requirements for all Machines.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American National Standards Institute (ANSI) Standards

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues.
ANSI B11.3 - Safety Requirements for the Construction, Care, and Use of Power Press Brakes.
ANSI/NFPA 79 - Metalworking Machine Tools and Plastics Machinery, Electrical Standards for.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

American Society For Testing and Materials (ASTM) Standards

ASTM D 3951 - Commercial Packaging, Standard Practice for.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification will take precedence.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.2, 6.2.1, and 6.3).

3.2 Design. The press brake shall be new and one of the manufacturer's current models capable of bending steel sheet and plate in accordance with the requirements herein. The press brakes shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Class 1, single speed. Class 1 press brakes shall have an operating speed of not less than the figure shown in table I in strokes per minute (SPM) for the full ram cycle.

3.2.2 Class 2, two speed. Class 2 press brakes shall have a high speed and a low speed not less than the figures shown in table I in strokes per minute for the full ram cycle.

3.2.3 Class 3, variable speed. Class 3 press brakes shall have an infinitely variable speed range as shown in table I. The range shall be from the smaller figure or less to the larger figure or more.

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3.2.4 Class 4, automatic cycle. Class 4 press brakes shall have an automatic ram cycle incorporating a high speed advance, a low pressing speed with full tonnage capacity through the bend, and a high speed return. These speeds in strokes per minute shall be not less than the figures shown in table I.

3.2.5 Safety and health requirements. All press brakes covered by this specification shall be constructed in conformance with the applicable requirements of ANSI B11.3. Covers, guards, or other safety devices shall be provided for all parts of the press brake that present safety hazards. The safety devices shall not interfere with the operation of the press brake. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All parts, components, mechanisms, and assemblies furnished on the press brake, whether or not specifically required herein, shall comply with all of the requirements of OSHA 29 CFR 1910.212 that are applicable to the press brake itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.6 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.7 Interchangeability. To provide for replacement of worn or damaged parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or press brake.

3.2.8 Construction. The press brake shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the press brake to its maximum rating and capacity without permanent distortion.

3.2.9 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.10 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the press brake. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.11 Reclaimed materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable.

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3.2.12 Energy efficiency. The press brake and all its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the Industry.

3.2.13 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the press brake. Such processes may be used only for enhancing surface finish and appearance.

3.2.14 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.2.15 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.2.16 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.2.17 Painting. Unless otherwise specified (see 6.2.1), the press brake shall be painted in accordance with the manufacturer's commercial practice, provided that all surfaces to be painted shall be cleaned of all foreign matter and not less than one coat of primer and one coat of finish color is applied.

3.2.18 Threads. All threaded parts used on the press brake and its related attachments and accessories shall conform to FED-STD-H28.

3.2.19 Gears. All gears shall be constructed from a material suitable for the intended purpose. All gears shall be machined and heat treated by a process that will impart the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.2.20 Bearings. Bearings for the mainshaft, intermediate shaft, and eccentric shaft shall be bronze and shall be the replaceable sleeve type. Bearings for all other rotating parts shall be anti-friction ball or roller type.

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3.2.21 T-Slots. All T-slots shall be in accordance with ANSI B5.1.

3.2.22 Marking of instruction plates, and indicating plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.2.23 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each press brake. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.2.24 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each press brake. The nameplate shall contain the information listed below. If the press brake is a special model, the model designation shall include the model of the basic standard press brake and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment code
Date of manufacture

3.2.25 Hour meter. Unless otherwise specified (see 6.2.1), the press brake shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the press brake. Upon reaching the maximum accumulative hours, the meter readout shall automatically revert to zero and continue to totalize time.

3.2.26 Electromagnetic interference control. When specified (see 6.2.1), press brakes furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

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3.3 Components. The press brakes covered herein shall consist essentially of the components described in 3.3.1 through 3.3.7.3. Any additional components necessary for the operation of the press brake shall be furnished to the manufacturer's standard design and specification.

3.3.1 Frame. The frame shall be of the open throat, gap design with a depth dimension in conformance with the applicable requirements of table I. The frame shall consist of side housings, a bed plate, and a crown member or housing plate spacers, all of steel plate construction, welded and stress relieved or interlocked into a single unit. Contoured stress points shall be utilized to avoid concentration of stresses at sharp corners. The frame shall have adequate strength and rigidity to withstand all forces encountered when bending sheet and plate of full capacity length and thickness in accordance with table I, for the press brake being procured. The frame shall support the ram, ram adjustment mechanisms, clutch, brake, flywheel, drive assemblies, operating controls, electrical system, and any additional components necessary to the operation of the press brake. Means shall be provided for securely fastening the frame to the floor or foundation.

3.3.2 Bed plate. The bed plate shall be a one piece steel plate or a one piece steel plate with a wide and deep bed platen welded to the main bed plate except for bearing shoes incorporated in the interlock design. The bed plate shall be welded and stress relieved or interlocked to the side frames. The top surface shall be slotted the full length of the bed to accommodate the manufacturer's standard die holders, filler blocks, or die tongues. Means shall be provided for adjusting and securely locking die holders in the slot. In addition, the top surface of the bed plate shall have four or more T-slots, at right angles to the tongue slot, for attaching short dies or fixtures. Means shall be provided for attaching gauges and fixtures to the bed. When specified (see 6.2.1), the front and rear face of the bed shall be machined and provided with a method for mounting wide angle support brackets or flanges. A die holder or filler block and a die with a mating punch having lengths equal to the die surface length specified in table I, for the press brake being procured, shall be furnished with each press brake. The type of die and punch to be furnished shall be as specified and as fully described (see 6.2.1).

3.3.3 Ram. The ram shall be constructed of one-piece steel plate with the pitman assembly connected by ball and socket joints or hardened swivel pins. The sockets may be integral with or attached to the ram. The ram shall be equipped with adjustable slides working against steel or self-lubricating non-metallic ways. The lower edge of the ram shall be slotted to receive the manufacturer's standard die tongues. A positive clamping device shall be incorporated to prevent die movement under load. When specified (see 6.2.1), the front and rear face of the ram shall be machined and provided with a method for mounting wide angle brackets or flanges.

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3.3.3.1 Ram adjustment. A ram adjustment shall be furnished that permits accurate tilt of not less than 1/8" per foot or parallel adjustment of the desired angle between the ram and bed. The amount of ram adjustment shall conform to the requirements of table I. Unless otherwise specified, the ram adjusting device shall be manually operated by a hand crank or similar device on press brakes up through 35 tons capacity. When specified (see 6.2.1), the ram adjusting device shall be by a separate electric motor drive on press brakes up through 35 tons capacity. A motorized ram adjustment shall be furnished on all press brakes rated above 35 tons. Press brakes having a motorized ram adjustment shall have a means to stop the ram when the limit of adjustment has been reached. Two ram position indicators shall be furnished with both the manual and motorized ram adjustments. The ram position indicators shall be located on the operators side of the press brake and shall read in thousandths of an inch.

3.3.3.2 Ram Counterbalance. Unless otherwise specified, a ram counterbalance shall be furnished on press brakes rated at 150 tons mid-stroke and larger having a bending capacity of 14 feet in length and longer. When specified (see 6.2.1), a counterbalance shall be furnished on press brakes rated at less than 150 tons at mid-stroke.

3.3.4 Flywheel. The press brake shall be furnished with a flywheel of an appropriate size, mass, and rim speed that will provide kinetic energy to make repetitive capacity bends as listed in table I, with a speed variation not to exceed 20 percent. The flywheel furnished shall be dynamically balanced.

3.3.5 Clutch and brake assembly. Unless otherwise specified, either a mechanical,air, or hydraulic type clutch and brake assembly shall be furnished to control the ram throughout the entire stroke. When only one type clutch and brake assembly is acceptable, the particular type required shall be as specified (see 6.2.1). When acti- vated, the clutch shall engage the flywheel and impart motion to the ram. Upon release, the clutch shall disengage from the flywheel and the brake shall automatically stop motion of the ram. The action of the clutch and brake assembly shall permit inching of the ram in addition to performing complete cycle operations. Unless otherwise specified, the operation of the clutch shall be controlled by dual palm buttons. When specified (see 6.2. 1), the clutch shall be controlled by either a foot treadle bar or a foot switch as specified.

3.3.6 Back gauges. Unless otherwise specified, the press brake shall be equipped with manually adjustable rear operated back gauges. When specified (see 6.2.1), the back gauges shall be operated from the front of the press brake. Unless otherwise specified, front-operated back gauges shall be manually operated. When specified (see 6.2.1), the front-operated back gauges shall be power operated.

3.3.7 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79. Each press brake shall

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draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit, having a fused safety disconnect switch or circuit breaker. The press brake shall be initially wired for operation on 230 volts. An identified terminal for grounding the press brake when installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating.

3.3.7.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure.

3.3.7.2 Solid state components. Solid state design shall be used throughout for electronic components.

3.3.7.3 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 115 volts.

3.4 Sizes and capacities. Unless otherwise specified (see 6.2.1), the press brake shall meet or exceed the applicable requirements of table I for the class and size press brake specified. The size designation of press brakes covered by this specification is a combination of the maximum forming tonnage at the bottom of the stroke and the length of the die surface in feet. The bending capacity is the maximum length and thickness of mild steel sheet and plate, with 60,000 p.s.i. tensile strength, which the press brake can, with one stoke, air bend into a 90° angle, over a V-die having an opening eight times the metal thickness for sheet and plate up to 1/2" or ten times the metal thickness for plate thicker than 1/2". The inside radius of the bend shall be 5/32 of the die opening.

3.5 Performance requirements. In addition to meeting the requirements of 3.4, the press brakes covered by this specification shall also be capable of forming other angular and channel shapes and multiple punching.

3.5.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.6 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract.

3.7 Standard equipment. Unless otherwise specified (see 6.2.1), the press brake shall be furnished with all standard equipment normally furnished by the manufacturer including one set of maintenance and repair tools normally furnished with the press brake.

3.8 Optional equipment. Optional equipment shall be furnished as specified and as fully described (see 6.2.1).

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TABLE I. Sizes and capacities.

Size	Tons		Length of die surface, feet	Bending capacity length x thickness	Distance between housings	Throat depth, inches	Stroke length, inches	Ram adjust, inches	Die space, stroke down, adjust up, inches	Motor H.P.	Speed (Strokes per minute)					
	Mid-stroke	Bottom of stroke									Class 1	Class 2	Class 3	Class 4		
														Advance	Press	Return
15-4	10	15	4	4' x 16 ga.	3'8"	6	2	1-1/2	7	1	-	-	20-45	-	-	-
15-6			6	6' x 18 ga.	5'2"											
15-10			10	10' x 24 ga.	8'6"											
25-6	17	25	6	6' x 16 ga.	5'2"	6	2	2	7	1-1/2	40	-	20-40	50	5	50
25-8			8	8' x 20 ga.	5'2"											
25-10			10	10' x 20 ga.	8'6"											
25-12			12	12' x 22 ga.	10'6"											
35-5	23	35	5	4' x 10 ga.	5'2"	8	3	3	9	2	40	-	-	-	-	-
35-8			8	8' x 16 ga.	5'2"											
55-6	35	55	6	6' x 10 ga.	4'6"	6	2-1/2	3	12	3	40	-	20-40	40	5	40
55-8			8	8' x 12 ga.	6'6"											
55-10			10	10' x 14 ga.	8'6"											
55-12			12	12' x 16 ga.	10'6"											
75-6	50	75	6	6' x 10 ga.	4'2"	7	2-1/2	4	12	3	40	-	20-40	40	5	40
75-8			8	8' x 10 ga.	6'6"											
75-10			10	10' x 12 ga.	8'6"											
75-12			12	12' x 14 ga.	10'6"											
75-14			14	14' x 14 ga.	12'6"											
75-16			16	16' x 16 ga.	14'6"											
75-18			18	18' x 16 ga.	16'6"											
75-20			20	20' x 18 ga.	18'6"											
75-22			22	22' x 20 ga.	20'6"											
90-6	60	90	6	6' x 3/16"	4'6"	8	3	4	12	5	35	-	20-50	35	6	35
90-8			8	8' x 3/16"	6'6"											
90-10			10	10' x 10 ga.	8'6"											
90-12			12	12' x 12 ga.	10'6"											
90-14			14	14' x 14 ga.	12'6"											
90-16			16	16' x 16 ga.	14'6"											
90-18			18	18' x 18 ga.	16'6"											
90-20			20	20' x 20 ga.	18'6"											
90-22			22	22' x 22 ga.	20'6"											
120-6	80	120	6	6' x 1/4"	4'6"	8	3	5	12	5	35	-	20-40	35	5	35
120-8			8	8' x 3/16"	6'6"											
120-10			10	10' x 3/16"	8'6"											
120-12			12	12' x 10 ga.	10'6"											
120-14			14	14' x 12 ga.	12'6"											
120-16			16	16' x 12 ga.	14'6"											
120-18			18	18' x 12 ga.	16'6"											
120-20			20	20' x 12 ga.	18'6"											
120-22			22	22' x 12 ga.	20'6"											
135-6	90	135	6	6' x 1/4"	4'5"	8	3	5	12	7	35	30 & 4	20-40	30	4	30
135-8			8	8' x 1/4"	6'5"											
135-10			10	10' x 3/16"	8'4"											
135-12			12	12' x 3/16"	10'5"											
135-14			14	14' x 10 ga.	12'5"											

NOTE: All dimensions shall be not less than those shown. Where a range is shown, the dimensions shall be not greater than the lowest figure to not less than the highest figure.

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TABLE I. Sizes and capacities. (cont'd)

Size	Tons		Length of die surface, feet	Bending capacity length x thickness	Distance between housings	Throat depth, inches	Stroke Length, inches	Ram adjust, inches	Die space, stroke down, adjust up, inches	Motor H.P.	Speed (Strokes per minute)					
	Mid-stroke	Bottom of stroke									Class 1	Class 2	Class 3	Class 4		
														Advance	Press	Return
150-6			6	6' x 5/16"	4'6"											
150-8			8	8' x 1/4"	6'6"											
150-10			10	10' x 3/16"	8'6"											
150-12			12	12' x 3/16"	10'6"											
150-14	100	150	14	14' x 10 ga.	12'6"	8	3	5	12	7	25	-	20-40	40	5	40
150-16			16	16' x 10 ga.	14'6"											
150-18			18	18' x 10 ga.	16'6"											
150-20			20	20' x 12 ga.	18'6"											
150-22			22	22' x 12 ga.	20'6"											
175-6			6	6' x 3/8"	4'6"											
175-8			8	8' x 5/16"	6'6"											
175-10			10	10' x 1/4"	8'6"											
175-12	116	175	12	12' x 3/16"	10'6"	8	3	5	12	10	30	30 & 4	20-40	30	4	30
175-14			14	14' x 3/16"	12'6"											
175-16			16	16' x 10 ga.	14'6"											
175-18			18	18' x 10 ga.	16'6"											
175-20			20	20' x 10 ga.	18'6"											
175-22			22	22' x 12 ga.	20'6"											
200-6			6	5' x 3/8"	4'5"											
200-8			8	8' x 5/16"	6'4"											
200-10	140	200	10	10' x 1/4"	8'5"	8	3	5	12	15	25	-	-	-	-	-
200-12			12	12' x 3/16"	10'5"											
225-6			6	6' x 3/8"	4'6"											
225-8			8	8' x 5/16"	6'6"											
225-10			10	10' x 1/4"	8'6"											
225-12	150	225	12	12' x 1/4"	10'6"	8	3	5	12	15	30	30 & 4	20-30	30	4	30
225-14			14	14' x 3/16"	12'6"											
225-16			16	16' x 3/16"	14'6"											
225-18			18	18' x 3/16"	16'6"											
225-20			20	20' x 3/16"	18'6"											
225-22			22	22' x 3/16"	20'6"											
275-8			8	8' x 3/8"	6'6"											
275-10			10	10' x 3/8"	8'6"											
275-12			12	12' x 3/8"	10'6"											
275-14	183	275	14	14' x 5/16"	12'6"	8	4	6	12	15	30	30 & 10	20-40	40	15	40
275-16			16	16' x 5/16"	14'6"											
275-18			18	18' x 1/4"	16'6"											
275-20			20	20' x 1/4"	18'6"											
275-22			22	20' x 1/4"	20'6"											
300-8			8	8' x 3/8"	6'6"											
300-10			10	10' x 3/8"	8'6"											
300-12	200	300	12	12' x 3/8"	10'6"	10	4	6	12	15	30	30 & 4	-	30	4	30
300-14			14	14' x 5/16"	12'6"											
300-16			16	16' x 1/4"	14'6"											
300-18			18	18' x 1/4"	16'6"											
300-20			20	20' x 1/4"	18'6"											

NOTE: All dimensions shall be not less than those shown. Where a range is shown, the dimensions shall be not greater than the lowest figure to not less than the highest figure.

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TABLE I. Sizes and capacities. (cont'd)

Size	Tons		Length of die surface, feet	Bending capacity length x thickness	Distance between housings	Throat depth, inches	Stroke Length, inches	Ram adjust, inches	Die space, stroke down, adjust up, inches	Motor H.P.	Speed (Strokes per minute)					
	Mid-stroke	Bottom of stroke									Class 1	Class 2	Class 3	Class 4		
														Advance	Press	Return
325-8	216	325	8	8' x 3/8"	6'6"	10	4	6	12	20	25	25 & 8	20-40	40	5	40
325-10			10	10' x 3/8"	8'6"											
325-12			12	12' x 5/16"	10'6"											
325-14			14	14' x 5/16"	12'6"											
325-16			16	16' x 5/16"	14'6"											
325-18			18	18' x 5/16"	16'6"											
325-20			20	20' x 1/4"	18'6"											
325-22			22	22' x 1/4"	20'6"											
400-10	260	400	10	10' x 5/8"	6'6"	10	4	6	12	20	25	25 & 8	8-25	30	5	30
400-12			12	12' x 3/8"	8'6"											
400-14			14	14' x 3/8"	10'6"											
400-16			16	16' x 3/8"	12'6"											
400-18			18	18' x 5/16"	14'6"											
400-20			20	20' x 1/4"	16'6"											
400-22			22	22' x 1/4"	18'6"											
500-10	330	500	10	10' x 1/2"	6'6"	14	4	6	14	25	25	25 & 8	8-25	40	5	40
500-12			12	12' x 3/8"	8'6"											
500-14			14	14' x 3/8"	10'6"											
500-16			16	16' x 3/8"	12'6"											
500-18			18	18' x 3/8"	14'6"											
500-20			20	18' x 5/16"	16'6"											
500-22			22	22' x 5/16"	18'6"											
500-24			24	24' x 5/16"	20'6"											
600-10	400	600	10	10' x 5/8"	6'6"	14	4	6	14	30	25	25 & 8	8-25	40	5	40
600-12			12	12' x 1/2"	8'6"											
600-14			14	14' x 1/2"	10'6"											
600-16			16	16' x 3/8"	12'6"											
600-18			18	18' x 3/8"	14'6"											
600-20			20	20' x 3/8"	16'6"											
600-22			22	22' x 5/16"	18'6"											
600-24			24	24' x 5/16"	20'6"											
750-14	500	750	14	14' x 5/8"	10'6"	14	4	6	14	40	23	23 & 7	8-25	40	5	40
750-16			16	16' x 1/2"	12'6"											
750-18			18	18' x 1/2"	14'6"											
750-20			20	20' x 1/2"	16'6"											
750-22			22	22' x 3/8"	18'6"											
750-24			24	24' x 3/8"	20'6"											
750-26			26	26' x 3/8"	22'6"											
1000-10	650	1000	10	10' x 3/4"	6'5"	14	4	6	14	40	23	23 & 6	-	-	-	-
1000-12			12	12' x 3/4"	8'5"											
1000-14			14	14' x 3/4"	10'5"											
1000-16			16	16' x 5/8"	12'5"											
1000-18			18	18' x 5/8"	14'5"											
1000-20			20	20' x 5/8"	16'5"											
1000-22			22	22' x 5/8"	18'5"											
1000-24			24	24' x 1/2"	20'5"											
1000-26			26	26' x 3/8"	22'5"											

NOTE: All dimensions shall be not less than those shown. Where a range is shown, the dimensions shall be not greater than the lowest figure to not less than the highest figure.

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3.9 Workmanship. Workmanship of the press brake and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.4 and all tests in 4.5. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.3 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of at least the examination in 4.4 and the test in 4.5.1. Failure of the item to pass the examination and test shall be cause for rejection.

4.4 Examination. The press brake and equipment shall be visually examined to determine compliance with the requirements of this specification.

4.4.1 Safety and health requirements. The press brake shall be examined to determine compliance with all OSHA requirements applicable to the press brake itself, compliance with applicable requirements of ANSI B11.3, Section 4, and for compliance with other safety and health requirements specified by the procuring activity (see 3.2.5, 6.2.1, and 6.4).

4.5 Tests.

4.5.1 Operational test. The press brake shall be operated at no load for not less than 30 minutes, during which time the controls, electric motors, tilting device, and ram shall be actuated and observed to determine compliance with the requirements specified herein.

4.5.2 Performance tests. The press brake shall be fitted with a punch and a V-die having an opening eight times the metal thickness for sheet and plate up to 1/2" and ten times the metal thickness for plate thicker than 1/2". The inside radius shall be 5/32 of the die opening. The punch and

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V-die shall be properly designed to compensate for material spring back. Each test sheet or plate shall be low carbon steel with a tensile strength of 60,000 p.s.i. The following tests shall be performed.

4.5.2.1 Full length bend test. The test piece shall be of the type and strength as stated in 4.5.2, having a length and thickness equal to the maximum capacity of the press brake as specified in table I. The test shall be accomplished by inserting the test piece between the dies and forming a 90° bend with one stroke of the ram. A total of 5 test bends shall be made, after which the press brake and the test pieces shall be examined. The press brake shall not be bowed, sprung, have chipped components or otherwise be distorted or broken. Each bend shall be clean and uniform throughout. The bend of each of the five test pieces shall be of the correct angle within plus or minus one and one- half degree for sheet up to 10 gage in thickness and plus or minus two and one-half degrees for plate stock over 10 gage in thickness.

4.5.2.2 Repetitive air bending test. The press brake shall be tested for accuracy in repetitive air bending by making air bends in pieces of test sheet or plate 6 inch x 6 inch or larger of the type and strength used in the full length bends. A test piece shall be positioned under each end of the ram and an air bend accomplished. This operation shall be repeated five times. Five additional test pieces shall be bent individually at the left end of the ram with no material under the opposite end. This operation shall be repeated for the right end of the ram. The twenty test pieces, when compared shall not vary more than plus or minus one and one-half degree for sheet up to 10 gage thickness and plus or minus two and one-half degrees for plate stock over 10 gage in thickness.

4.6 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Packaging, packing, and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A packaging, level A or level B packing and special marking shall be accomplished in accordance with MIL-M-18058.

5.2 Marking.

5.2.1 Civil agencies. When specified (see 6.2.1), additional marking in accordance with FED-STD-123 shall be furnished.

6. NOTES

6.1 Intended use. The press brakes covered by this specification are intended for use in forming, bending, flanging, multiple punching and similar operations on sheet metal and plate.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Purchasers should specify their requirements in acquisition documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.

b. Class and size of machine required (see 1.2 and table I).

c. First article approval, if required (see 3.1).

d. Specify additional safety and health requirements (see 3.2.5, 4.4.1 and 6.4).

e. Specific type measurement system, if required; specify either "U.S. Customary (US)" or "International System (SI), (see 3.2.10).

f. Painting, if different (see 3.2.17).

g. Lubrication chart or plate, if different (see 3.2.23).

h. Nameplate, if different (see 3.2.24).

i. Hour meter, if different (see 3.2.25).

j. If electromagnetic interference control is required, specify the equipment and subsystem class and the emmision and susceptibility required (see 3.2.26).

k. Machined bed, if required (see 3.3.2).

l. Type of die and punch required; fully describe (see 3.3.2).

m. Machined ram, if required (see 3.3.3).

n. Ram adjustment, if different (see 3.3.3.1).

o. Ram counterbalance, if different (see 3.3.3.2).

p. If the clutch and brake assembly is required to be a specific type, state required type (see 3.3.5).

q. Operating control, specify type (see 3.3.5).

r. Back gauges, if different (see 3.3.6).

s. Front-operated back gauges, if different (see 3.3.6).

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t. Electrical equipment, if different (see 3.3.7).

u. Motor and enclosure, if different (see 3.3.7.1).

v. Sizes and capacities, if different (see 3.4 and table I).

w. Specify additional performance requirements (see 3.5.1).

x. Standard equipment, if different (see 3.7).

y. Optional equipment, if required, fully described (see 3.8).

z. First article inspection, if different (see 4.2).

aa. Quality conformance inspection, if different (see 4.3).

bb. Level of packaging, packing and marking, if different (see 5.1).

cc. When marking is required to conform to FED-STD-123 (see 5.2.1).

6.2.2 Contract data requirements. Required technical data such as operators manuals, parts lists, wiring diagrams, examination, original alignment and tolerance test reports, and other instructions for operation and maintenance (as identified on a DD Form 1664, Data Item Description) shall be specified on a DD Form 1423, Contract Data Requirements List incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer shall include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the press brake itself. It does not require compliance with those OSHA requirements that concern "the press brake in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual press brakes in the environment) the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional press brake requirements that will integrate the new press brake into its future operating environment. The press brake shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial press brakes supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required, as in most cases, the guard configuration is dependent on the size and configuration of

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the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1(d).

6.5 Cross reference of classification changes: Classes previously included in 00-P-631C have been maintained in this revision. Size changes from the previous issue of this specification are as follows:

00-P-631D	00-P-631D (contd)	00-P-631D (contd)
35-5 (new size)	175-20 (new size)	325-16 (new size)
35-8 (new size)	175-22 (new size)	325-18 (new size)
75-14 (new size)	200-6 (new size)	325-20 (new size)
75-16 (new size)	200-8 (new size)	325-22 (new size)
75-18 (new size)	200-10 (new size)	400-20 (new size)
75-20 (new size)	200-12 (new size)	400-22 (new size)
75-22 (new size)	225-18 (new size)	500-20 (new size)
90-18 (new size)	225-20 (new size)	500-22 (new size)
90-20 (new size)	225-22 (new size)	500-24 (new size)
90-22 (new size)	275-8 (new size)	600-20 (new size)
120-16 (new size)	275-10 (new size)	600-22 (new size)
120-18 (new size)	275-12 (new size)	600-24 (new size)
135-16 deleted	275-14 (new size)	750-10 deleted
120-22 (new size)	275-16 (new size)	750-12 deleted
150-6 (new size)	275-18 (new size)	750-20 (new size)
150-8 (new size)	275-20 (new size)	750-22 (new size)
150-10 (new size)	275-22 (new size)	750-24 (new size)
150-12 (new size)	300-18 (new size)	750-26 (new size)
150-14 (new size)	300-20 (new size)	1000-20 (new size)
150-16 (new size)	325-8 (new size)	1000-22 (new size)
150-18 (new size)	325-10 (new size)	1000-24 (new size)
150-20 (new size)	325-12 (new size)	1000-26 (new size)
150-22 (new size)	325-14 (new size)	

MILITARY INTERESTS:

Custodians

Army - AL
Navy - SH
Air Force - 99

Review Activities

Army - AL, ER
Navy - SH
Air Force - 84
DGSC - GS

User Activities

Army - ME
Navy - MC, YD

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

PREPARING ACTIVITY:

DLA-IP

Project 3441-0111

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
00-P-631D	Press Brake, Power Operated, Mechanical, Sheet and Plate

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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